Causal Analysis Report

BPI Challenge 2017 - Offer log.xes.gz-dfg

Decision points:	108033
Choice sets:	4
Transitions fitted:	9
Significant (p<0.05):	0
Generated:	2026-06-09 09:12

Step 1 — Association test (logistic regression + permutation)
H_0: choice ⊥ util | choice_set.   reject_H0=True → util predicts routing.

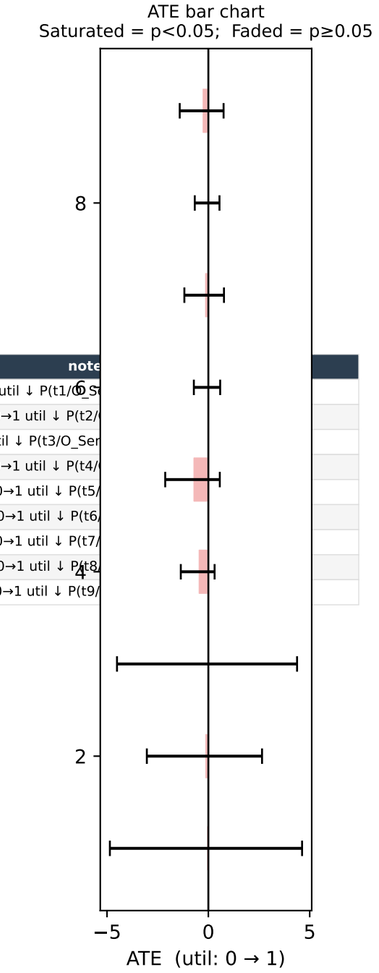
choice_set	choice_set_labels	n	choices_seen	coef	ci_95	perm_p_value	reject_H0	note
(1, 2, 3)	('O_Sent (online only)_t1', 'O_Cancelled_t2', 'O_Sent (mail and online)_t3')	42936	1, 2, 3	0.1861	[-0.050, 0.687]	0.453	False	higher util → t3 (O_Sent (mail and online)_t3) over t1 (O_Sent (online only)_t1)
(4, 5)	('O_Cancelled_t4', 'O_Returned_t5')	1908	4, 5	-14.5507	[-15.610, -13.393]	0.0	True	higher util → t4 (O_Cancelled_t4) over t5 (O_Returned_t5)
(6, 7)	('O_Cancelled_t6', 'O_Returned_t7')	38637	6, 7	-21.3542	[-22.029, -20.711]	0.0	True	higher util → t6 (O_Cancelled_t6) over t7 (O_Returned_t7)
(8, 9)	('O_Cancelled_t8', 'O_Accepted_t9')	19683	8, 9	-24.0611	[-25.141, -23.065]	0.0	True	higher util → t8 (O_Cancelled_t8) over t9 (O_Accepted_t9)

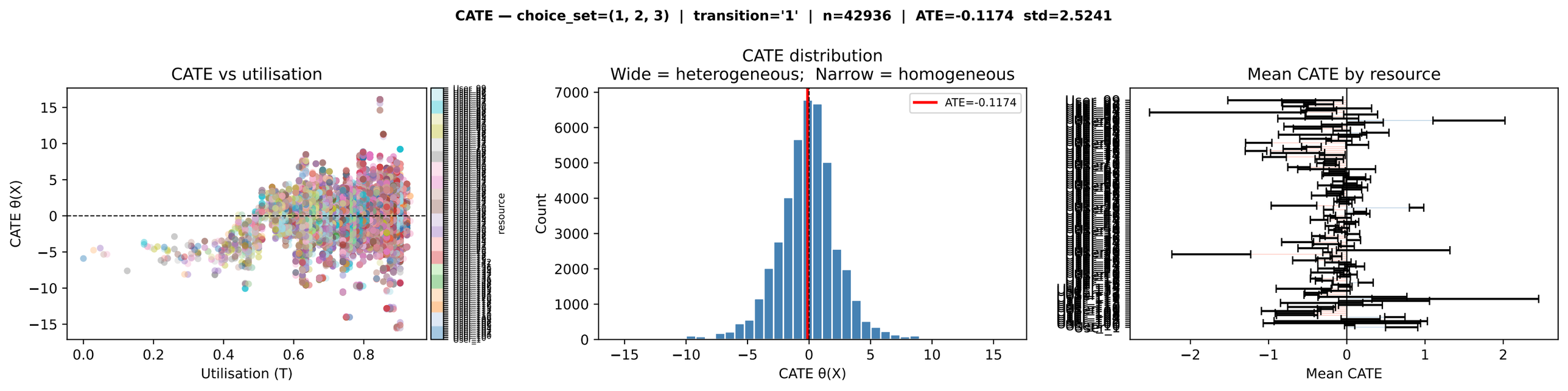
Step 2 — Double ML (partially linear regression, HC3 robust SE)
θ = linear effect of 1-unit ↑ util on P(chose k), after partialling out X.
Equals ATE when the effect is homogeneous. Otherwise: variance-weighted average of CATEs.

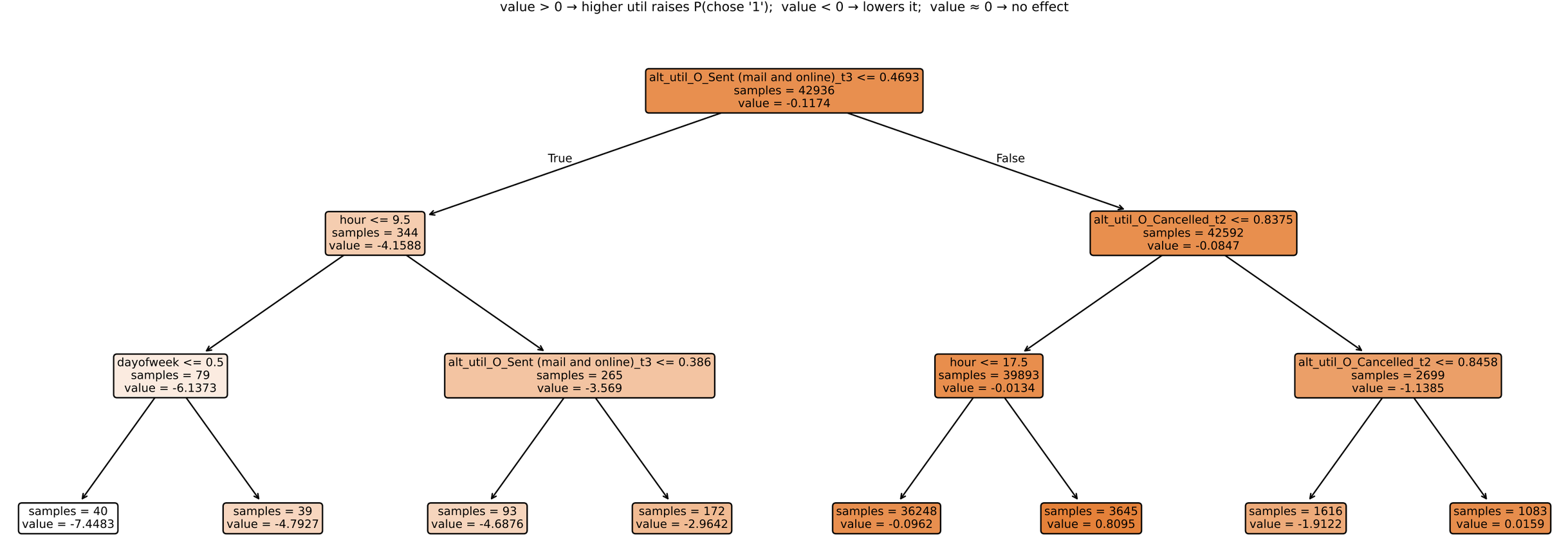
choice_set	choice_set_labels	n	choices_seen	transition	transition_label	alt_utils	theta	se	ci_95	p_value	reject_H0	note
(1, 2, 3)	('O_Sent (online only)_t1', 'O_Cancelled_t2', 'O_Sent (mail and online)_t3')	42936	1, 2, 3	1	O_Sent (online only)_t1	O_Cancelled_t2, O_Sent (mail and online)_t3	0.133286	0.105961	[-0.0744, 0.3410]	0.2084	False	θ=+0.1333: 1-unit ↑ util ↑ P(t1/O_Sent (online only)_t1) by 13.3%
(1, 2, 3)	('O_Sent (online only)_t1', 'O_Cancelled_t2', 'O_Sent (mail and online)_t3')	42936	1, 2, 3	2	O_Cancelled_t2	O_Sent (mail and online)_t3, O_Sent (online only)_t1	0.002628	0.055393	[-0.1059, 0.1112]	0.9622	False	θ=+0.0026: 1-unit ↑ util ↑ P(t2/O_Cancelled_t2) by 0.3%
(1, 2, 3)	('O_Sent (online only)_t1', 'O_Cancelled_t2', 'O_Sent (mail and online)_t3')	42936	1, 2, 3	3	O_Sent (mail and online)_t3	O_Cancelled_t2, O_Sent (online only)_t1	0.140065	0.103935	[-0.0636, 0.3438]	0.1778	False	θ=+0.1401: 1-unit ↑ util ↑ P(t3/O_Sent (mail and online)_t3) by 14.0%
(4, 5)	('O_Cancelled_t4', 'O_Returned_t5')	1908	4, 5	4	O_Cancelled_t4	O_Returned_t5	-0.589346	0.112928	[-0.8107, -0.3680]	0.0	True	θ=-0.5893: 1-unit ↑ util ↓ P(t4/O_Cancelled_t4) by 58.9%
(4, 5)	('O_Cancelled_t4', 'O_Returned_t5')	1908	4, 5	5	O_Returned_t5	O_Cancelled_t4	-0.419049	0.17297	[-0.7581, -0.0800]	0.0154	True	θ=-0.4190: 1-unit ↑ util ↓ P(t5/O_Returned_t5) by 41.9%
(6, 7)	('O_Cancelled_t6', 'O_Returned_t7')	38637	6, 7	6	O_Cancelled_t6	O_Returned_t7	-0.080201	0.012671	[-0.1050, -0.0554]	0.0	True	θ=-0.0802: 1-unit ↑ util ↓ P(t6/O_Cancelled_t6) by 8.0%
(6, 7)	('O_Cancelled_t6', 'O_Returned_t7')	38637	6, 7	7	O_Returned_t7	O_Cancelled_t6	-0.077826	0.02057	[-0.1181, -0.0375]	0.0002	True	θ=-0.0778: 1-unit ↑ util ↓ P(t7/O_Returned_t7) by 7.8%
(8, 9)	('O_Cancelled_t8', 'O_Accepted_t9')	19683	8, 9	8	O_Cancelled_t8	O_Accepted_t9	-0.057061	0.028713	[-0.1133, -0.0008]	0.0469	True	θ=-0.0571: 1-unit ↑ util ↓ P(t8/O_Cancelled_t8) by 5.7%
(8, 9)	('O_Cancelled_t8', 'O_Accepted_t9')	19683	8, 9	9	O_Accepted_t9	O_Cancelled_t8	-0.029327	0.039722	[-0.1072, 0.0485]	0.4603	False	θ=-0.0293: 1-unit ↑ util ↓ P(t9/O_Accepted_t9) by 2.9%

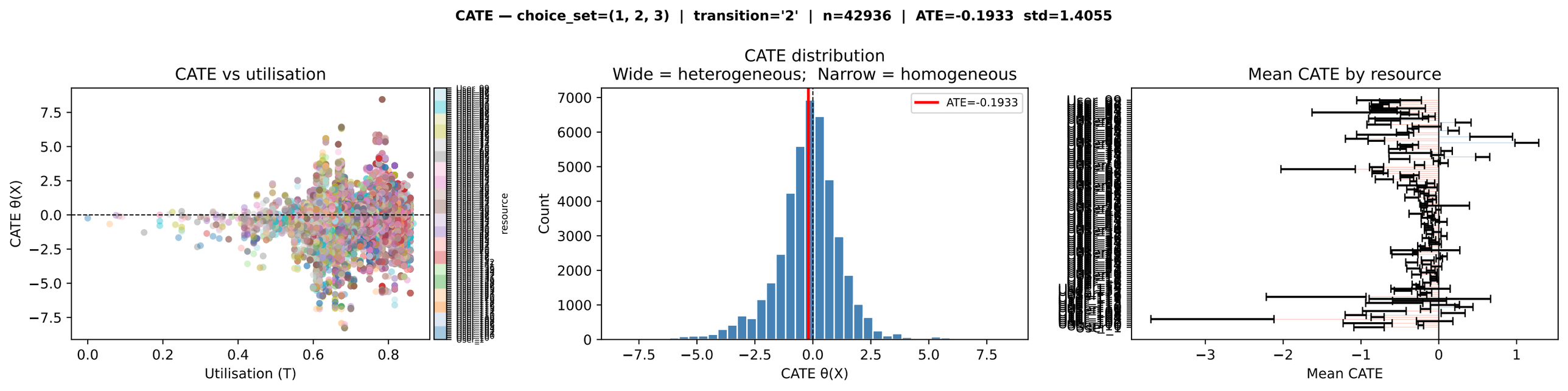
Step 3 — Causal Forest DML (ATE per transition)
ATE = mean of per-sample CATEs. Green rows = H0 rejected (p<0.05).
Less powerful for significance than Double ML, but basis for heterogeneity analysis.

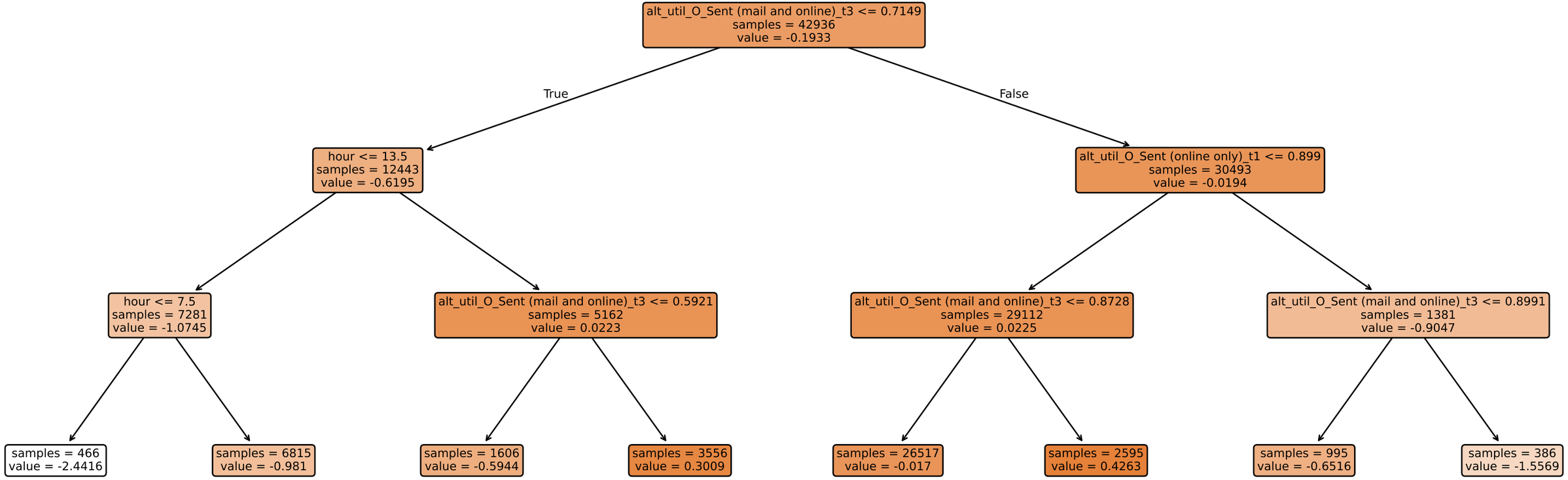
choice_set	choice_set_labels	n	choices_seen	transition	transition_label	alt_utils	ate	ci_95	p_value	reject_H0	note
(1, 2, 3)	('O_Sent (online only)_t1', 'O_Cancelled_t2', 'O_Sent (mail and online)_t3')	42936	1, 2, 3	1	O_Sent (online only)_t1	O_Cancelled_t2, O_Sent (mail and online)_t3	-0.117371	[-4.8594, 4.6247]	0.9613	False	ATE=-0.1174: 0→1 util ↓ P(t1/O_S
(1, 2, 3)	('O_Sent (online only)_t1', 'O_Cancelled_t2', 'O_Sent (mail and online)_t3')	42936	1, 2, 3	2	O_Cancelled_t2	O_Sent (mail and online)_t3, O_Sent (online only)_t1	-0.193329	[-3.0354, 2.6488]	0.8939	False	ATE=-0.1933: 0→1 util ↓ P(t2/(
(1, 2, 3)	('O_Sent (online only)_t1', 'O_Cancelled_t2', 'O_Sent (mail and online)_t3')	42936	1, 2, 3	3	O_Sent (mail and online)_t3	O_Cancelled_t2, O_Sent (online only)_t1	-0.065195	[-4.5076, 4.3773]	0.9771	False	ATE=-0.0652: 0→1 util ↓ P(t3/O_Sen
(4, 5)	('O_Cancelled_t4', 'O_Returned_t5')	1908	4, 5	4	O_Cancelled_t4	O_Returned_t5	-0.525971	[-1.3597, 0.3078]	0.2163	False	ATE=-0.5260: 0→1 util ↓ P(t4/(
(4, 5)	('O_Cancelled_t4', 'O_Returned_t5')	1908	4, 5	5	O_Returned_t5	O_Cancelled_t4	-0.78197	[-2.1284, 0.5645]	0.255	False	ATE=-0.7820: 0→1 util ↓ P(t5/(
(6, 7)	('O_Cancelled_t6', 'O_Returned_t7')	38637	6, 7	6	O_Cancelled_t6	O_Returned_t7	-0.066281	[-0.7155, 0.5829]	0.8414	False	ATE=-0.0663: 0→1 util ↓ P(t6/(
(6, 7)	('O_Cancelled_t6', 'O_Returned_t7')	38637	6, 7	7	O_Returned_t7	O_Cancelled_t6	-0.206431	[-1.1837, 0.7708]	0.6789	False	ATE=-0.2064: 0→1 util ↓ P(t7/(
(8, 9)	('O_Cancelled_t8', 'O_Accepted_t9')	19683	8, 9	8	O_Cancelled_t8	O_Accepted_t9	-0.06089	[-0.6704, 0.5487]	0.8448	False	ATE=-0.0609: 0→1 util ↓ P(t8/(
(8, 9)	('O_Cancelled_t8', 'O_Accepted_t9')	19683	8, 9	9	O_Accepted_t9	O_Cancelled_t8	-0.330853	[-1.4131, 0.7514]	0.549	False	ATE=-0.3309: 0→1 util ↓ P(t9/(

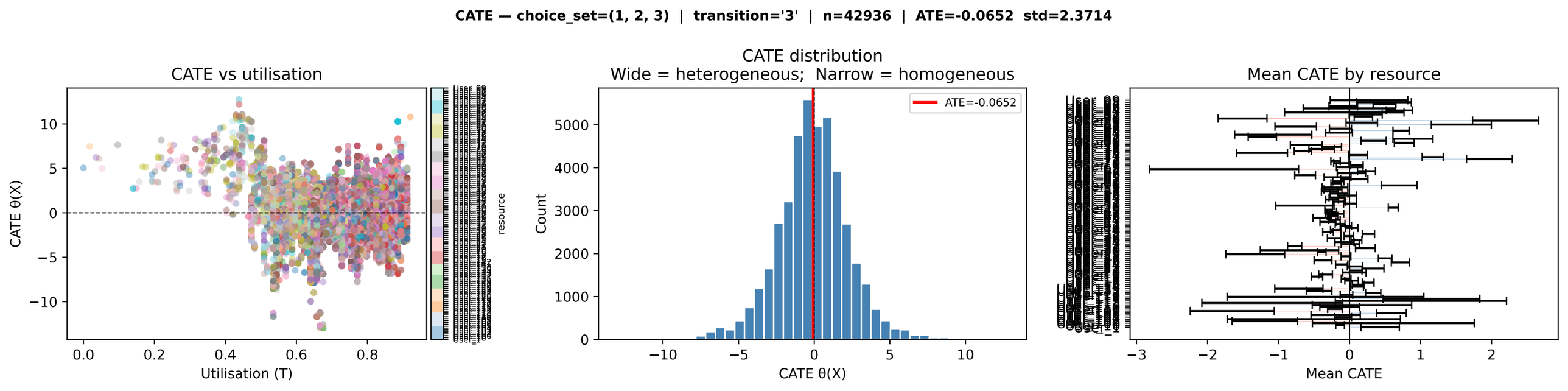


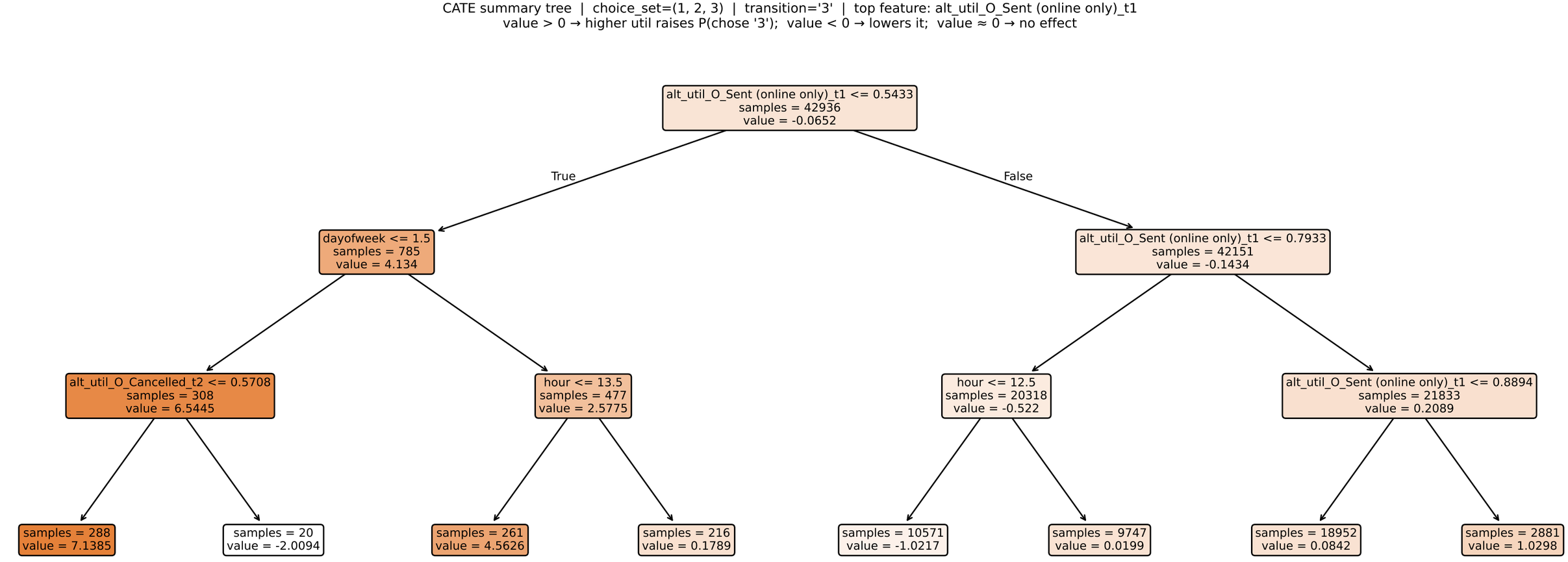

CATE — choice_set=(1, 2, 3) | transition='1' | n=42936 | ATE=-0.1174 std=2.5241
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
Count
CATE θ(X)
ATE=-0.1174
Mean CATE by resource
Mean CATE

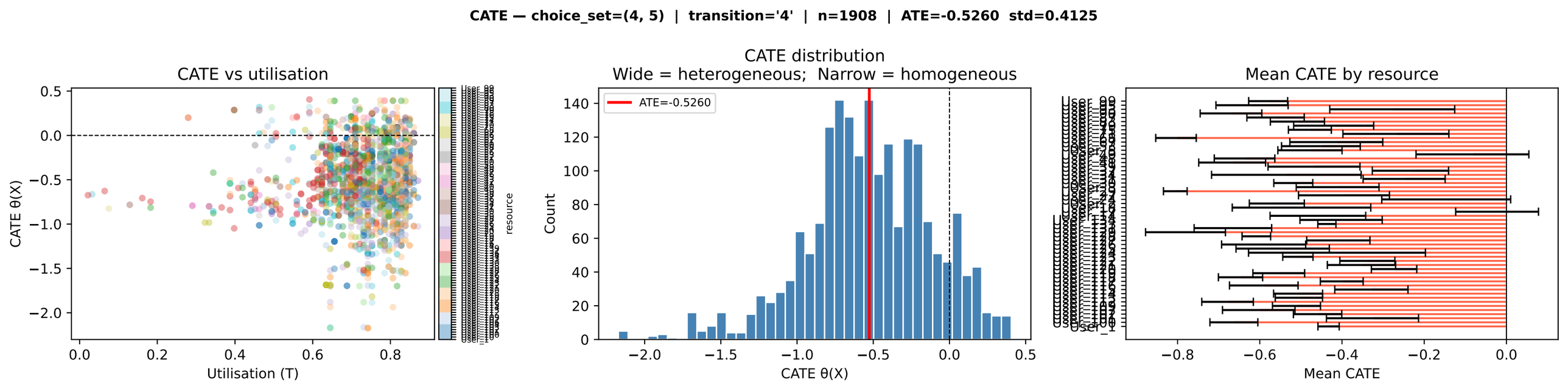

CATE summary tree | choice_set=(1, 2, 3) | transition='1' | top feature: alt_util_O_Sent (mail and online)_t3
value > 0 → higher util raises P(chose '1'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_O_Sent (mail and online)_t3 <= 0.4693
samples = 42936
value = -0.1174
True
False
hour <= 9.5
samples = 344
value = -4.1588
alt_util_O_Cancelled_t2 <= 0.8375
samples = 42592
value = -0.0847
dayofweek <= 0.5
samples = 79
value = -6.1373
alt_util_O_Sent (mail and online)_t3 <= 0.386
samples = 265
value = -3.569
hour <= 17.5
samples = 39893
value = -0.0134
alt_util_O_Cancelled_t2 <= 0.8458
samples = 2699
value = -1.1385
samples = 40
value = -7.4483
samples = 39
value = -4.7927
samples = 93
value = -4.6876
samples = 172
value = -2.9642
samples = 36248
value = -0.0962
samples = 3645
value = 0.8095
samples = 1616
value = -1.9122
samples = 1083
value = 0.0159

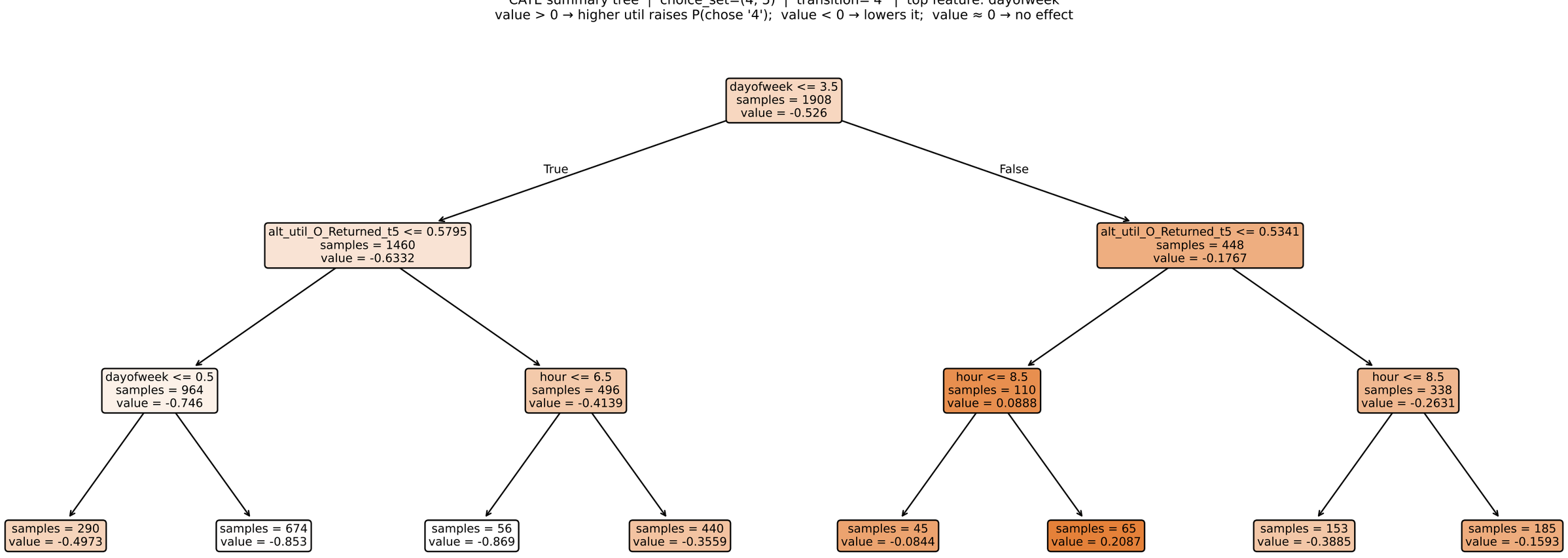

CATE — choice_set=(1, 2, 3) | transition='2' | n=42936 | ATE=-0.1933 std=1.4055
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.1933
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

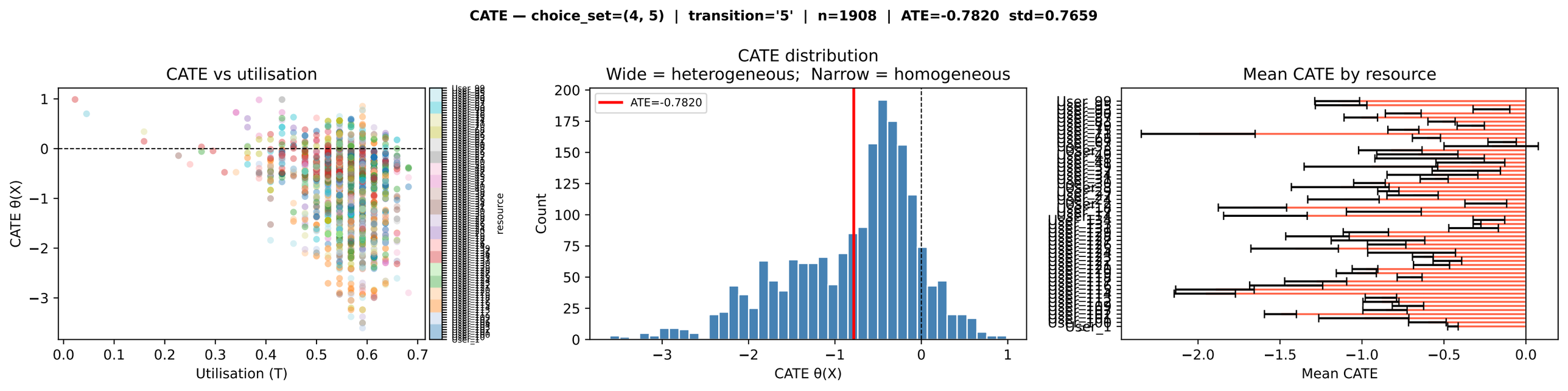

CATE summary tree | choice_set=(1, 2, 3) | transition='2' | top feature: alt_util_O_Sent (mail and online)_t3
value > 0 → higher util raises P(chose '2'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_O_Sent (mail and online)_t3 <= 0.7149
samples = 42936
value = -0.1933
True
False
hour <= 13.5
samples = 12443
value = -0.6195
alt_util_O_Sent (online only)_t1 <= 0.899
samples = 30493
value = -0.0194
hour <= 7.5
samples = 7281
value = -1.0745
alt_util_O_Sent (mail and online)_t3 <= 0.5921
samples = 5162
value = 0.0223
alt_util_O_Sent (mail and online)_t3 <= 0.8728
samples = 29112
value = 0.0225
alt_util_O_Sent (mail and online)_t3 <= 0.8991
samples = 1381
value = -0.9047
samples = 466
value = -2.4416
samples = 6815
value = -0.981
samples = 1606
value = -0.5944
samples = 3556
value = 0.3009
samples = 26517
value = -0.017
samples = 2595
value = 0.4263
samples = 995
value = -0.6516
samples = 386
value = -1.5569

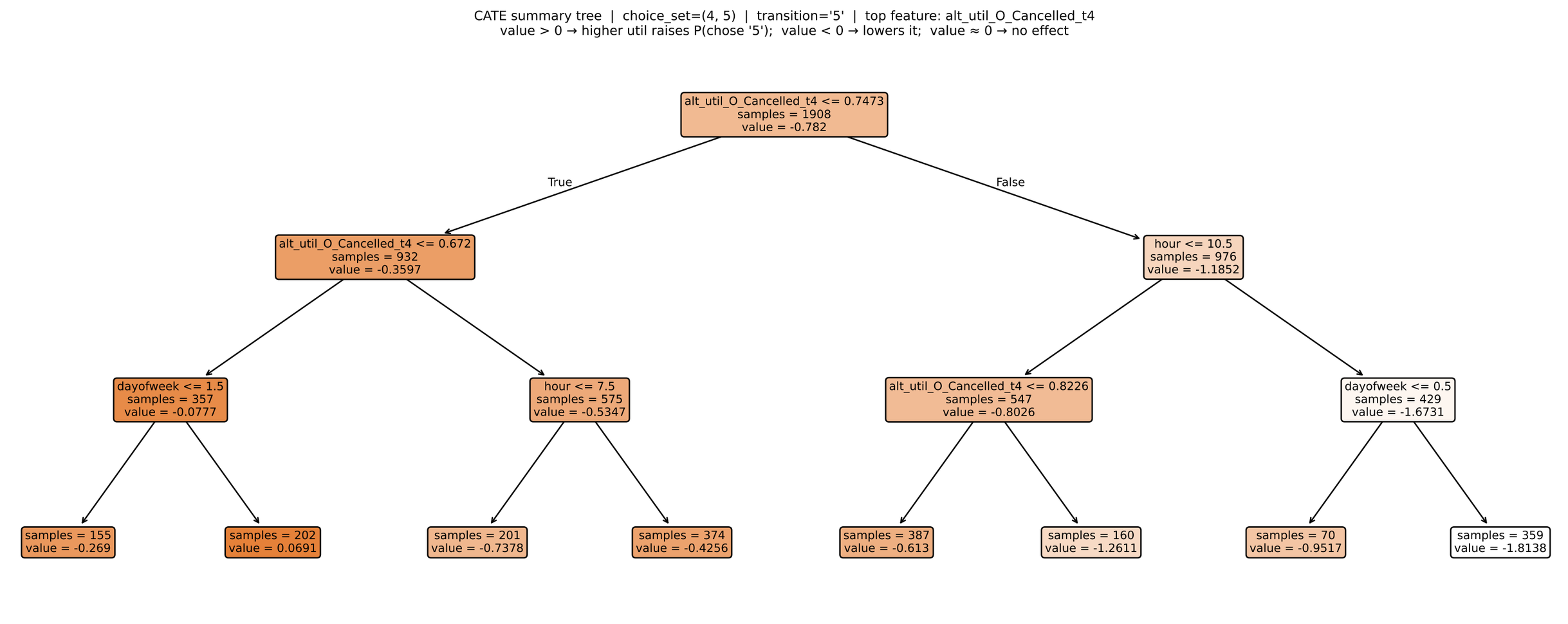

CATE — choice_set=(1, 2, 3) | transition='3' | n=42936 | ATE=-0.0652 std=2.3714
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.0652
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

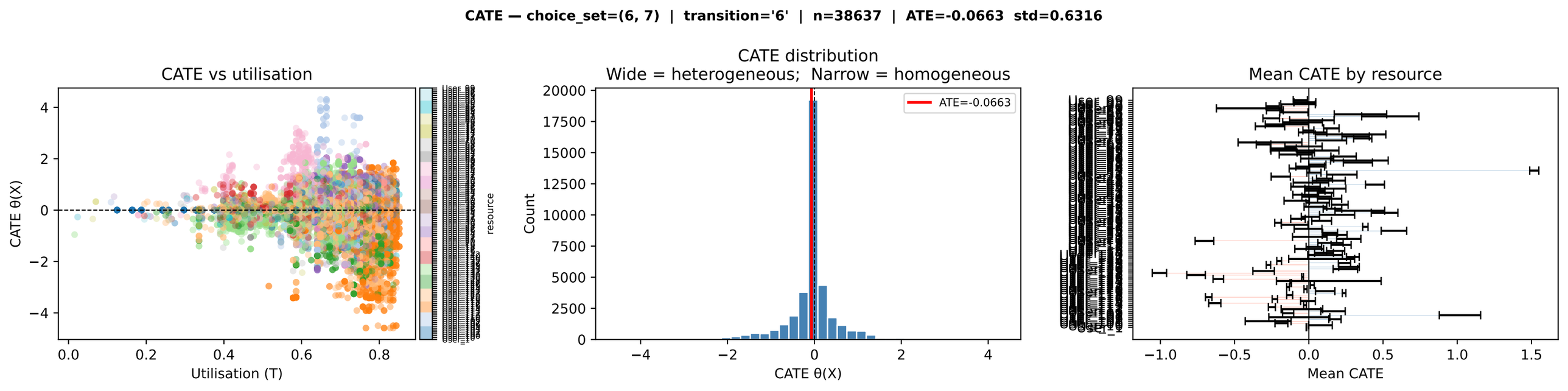

CATE summary tree | choice_set=(1, 2, 3) | transition='3' | top feature: alt_util_O_Sent (online only)_t1
value > 0 → higher util raises P(chose '3'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_O_Sent (online only)_t1 <= 0.5433
samples = 42936
value = -0.0652
True
False
dayofweek <= 1.5
samples = 785
value = 4.134
alt_util_O_Sent (online only)_t1 <= 0.7933
samples = 42151
value = -0.1434
alt_util_O_Cancelled_t2 <= 0.5708
samples = 308
value = 6.5445
hour <= 13.5
samples = 477
value = 2.5775
hour <= 12.5
samples = 20318
value = -0.522
alt_util_O_Sent (online only)_t1 <= 0.8894
samples = 21833
value = 0.2089
samples = 288
value = 7.1385
samples = 20
value = -2.0094
samples = 261
value = 4.5626
samples = 216
value = 0.1789
samples = 10571
value = -1.0217
samples = 9747
value = 0.0199
samples = 18952
value = 0.0842
samples = 2881
value = 1.0298

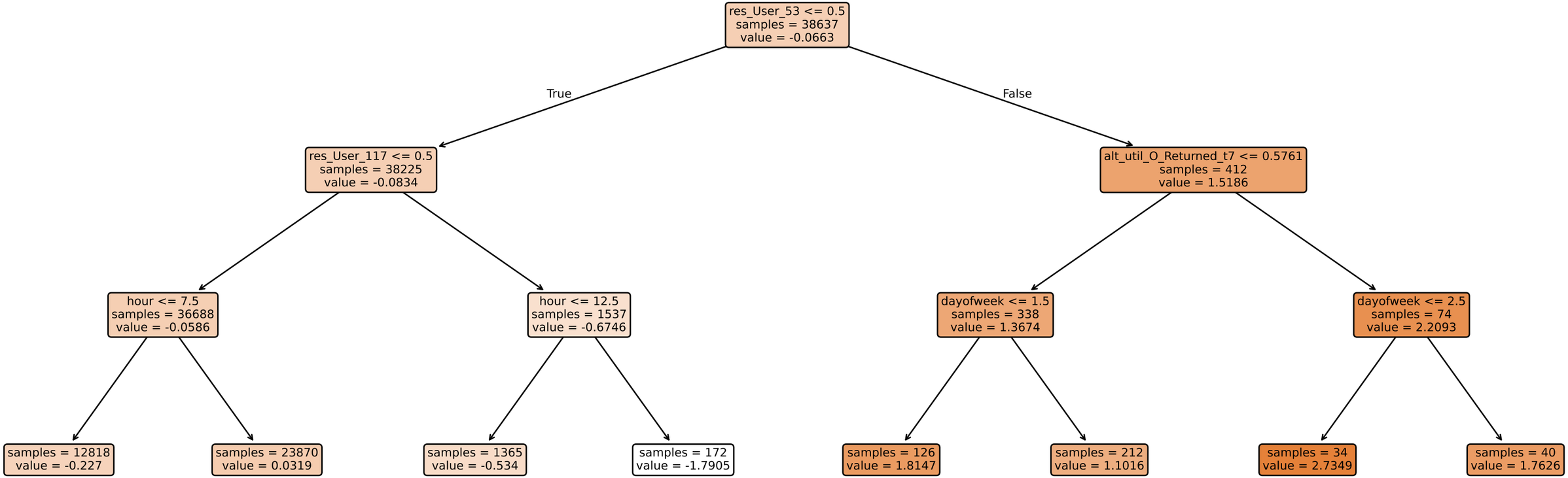

CATE — choice_set=(4, 5) | transition='4' | n=1908 | ATE=-0.5260 std=0.4125
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.5260
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

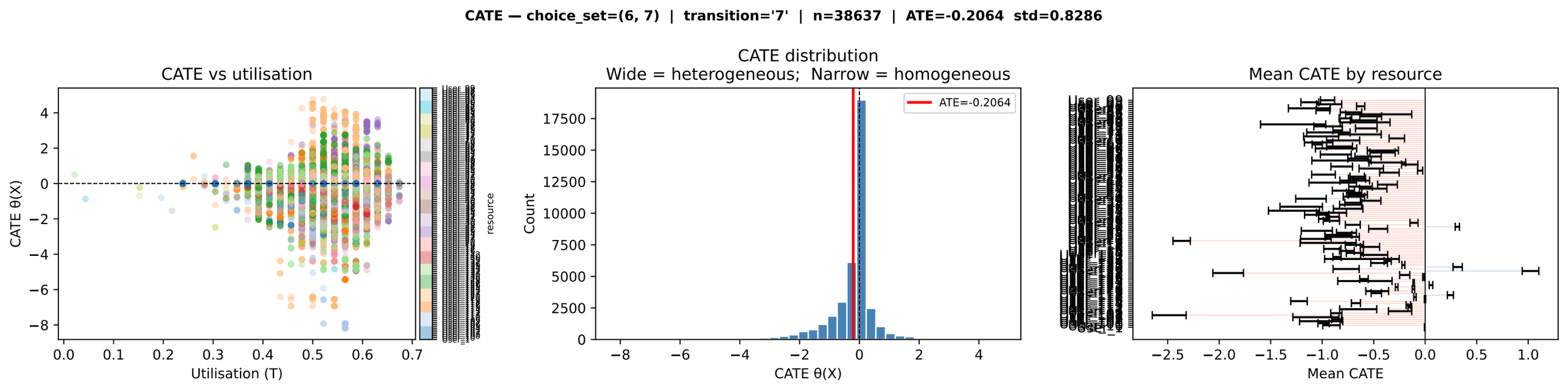

CATE summary tree | choice_set=(4, 5) | transition='4' | top feature: dayofweek
value > 0 → higher util raises P(chose '4'); value < 0 → lowers it; value ≈ 0 → no effect
dayofweek <= 3.5
samples = 1908
value = -0.526
True
False
alt_util_O_Returned_t5 <= 0.5795
samples = 1460
value = -0.6332
alt_util_O_Returned_t5 <= 0.5341
samples = 448
value = -0.1767
dayofweek <= 0.5
samples = 964
value = -0.746
hour <= 6.5
samples = 496
value = -0.4139
hour <= 8.5
samples = 110
value = 0.0888
hour <= 8.5
samples = 338
value = -0.2631
samples = 290
value = -0.4973
samples = 674
value = -0.853
samples = 56
value = -0.869
samples = 440
value = -0.3559
samples = 45
value = -0.0844
samples = 65
value = 0.2087
samples = 153
value = -0.3885
samples = 185
value = -0.1593

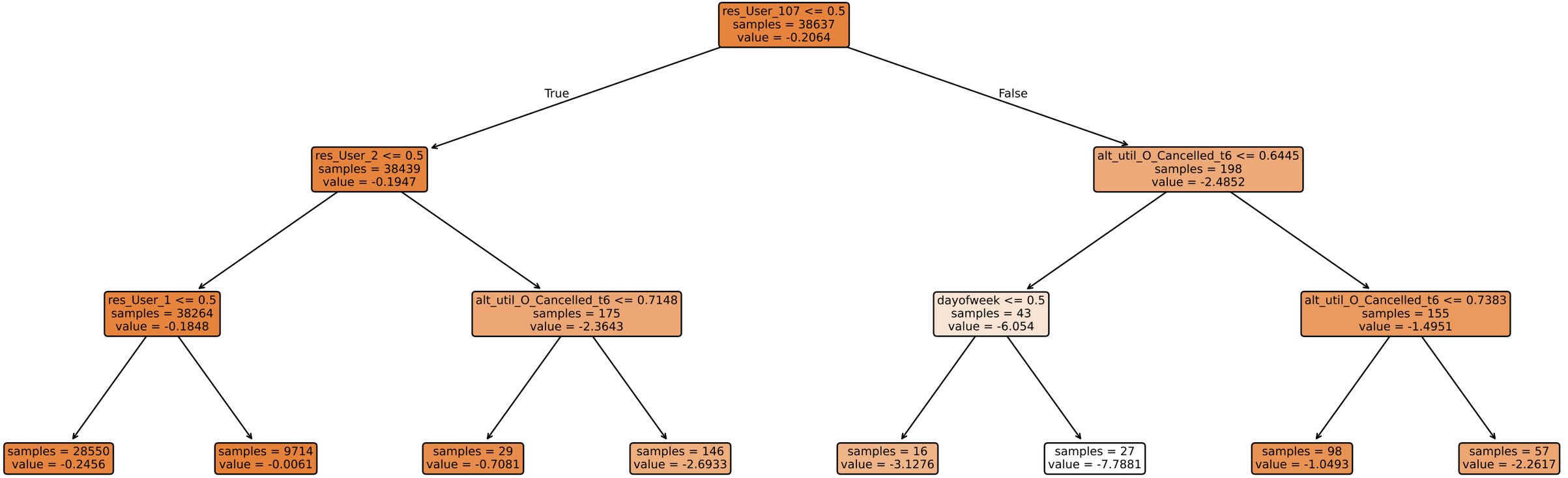

CATE — choice_set=(4, 5) | transition='5' | n=1908 | ATE=-0.7820 std=0.7659
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.7820
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

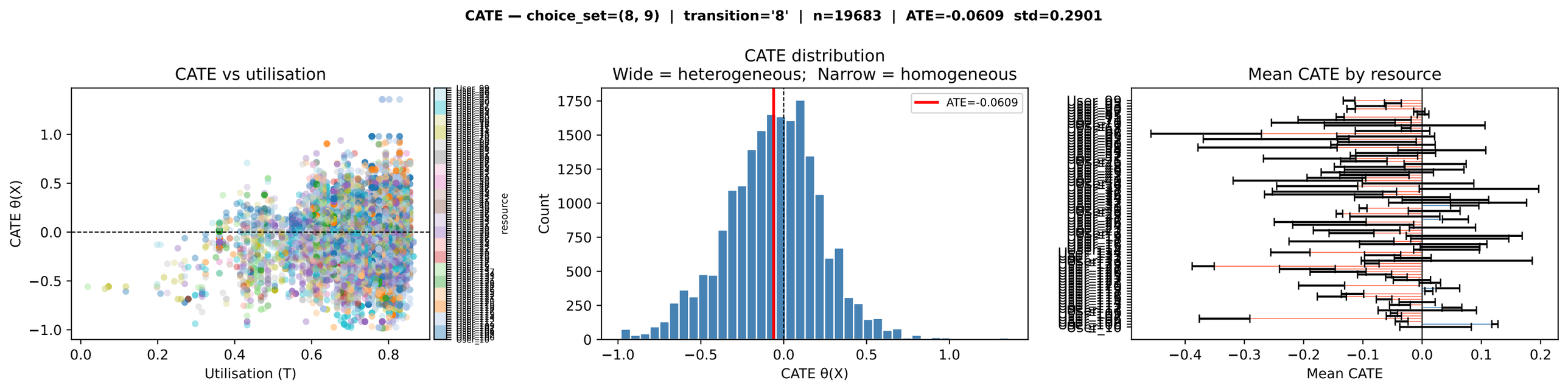

CATE summary tree | choice_set=(4, 5) | transition='5' | top feature: alt_util_O_Cancelled_t4
value > 0 → higher util raises P(chose '5'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_O_Cancelled_t4 <= 0.7473
samples = 1908
value = -0.782
True
False
alt_util_O_Cancelled_t4 <= 0.672
samples = 932
value = -0.3597
hour <= 10.5
samples = 976
value = -1.1852
dayofweek <= 1.5
samples = 357
value = -0.0777
hour <= 7.5
samples = 575
value = -0.5347
alt_util_O_Cancelled_t4 <= 0.8226
samples = 547
value = -0.8026
dayofweek <= 0.5
samples = 429
value = -1.6731
samples = 155
value = -0.269
samples = 202
value = 0.0691
samples = 201
value = -0.7378
samples = 374
value = -0.4256
samples = 387
value = -0.613
samples = 160
value = -1.2611
samples = 70
value = -0.9517
samples = 359
value = -1.8138

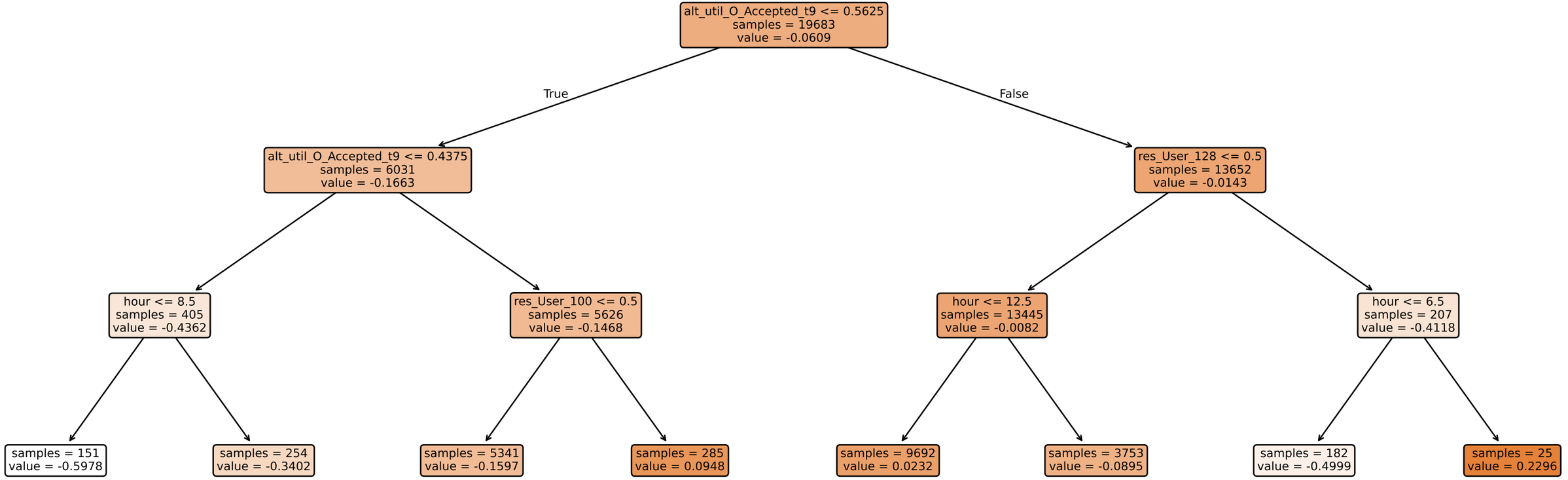

CATE — choice_set=(6, 7) | transition='6' | n=38637 | ATE=-0.0663 std=0.6316
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.0663
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

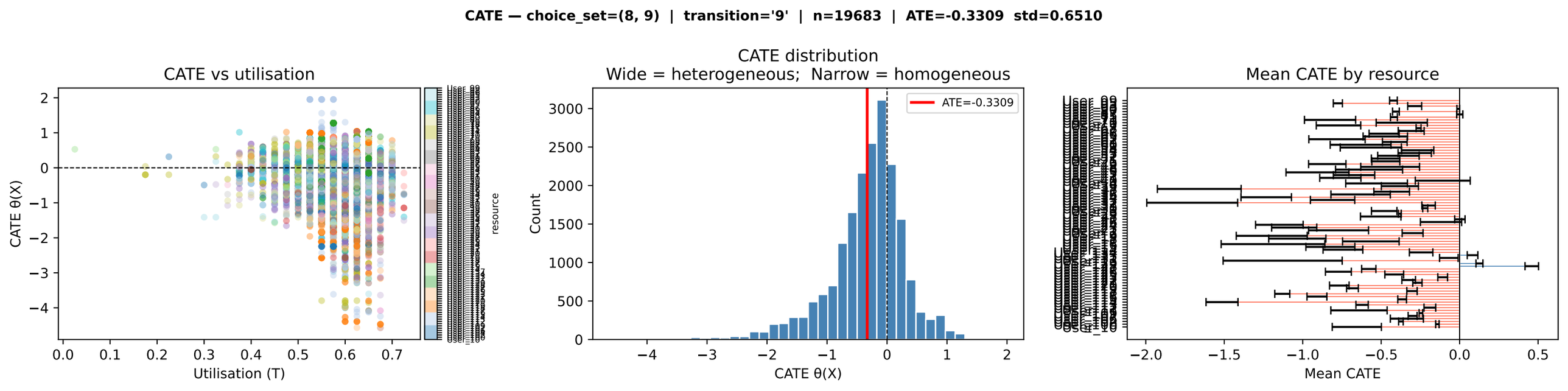

CATE summary tree | choice_set=(6, 7) | transition='6' | top feature: res_User_53
value > 0 → higher util raises P(chose '6'); value < 0 → lowers it; value ≈ 0 → no effect
res_User_53 <= 0.5
samples = 38637
value = -0.0663
True
False
res_User_117 <= 0.5
samples = 38225
value = -0.0834
alt_util_O_Returned_t7 <= 0.5761
samples = 412
value = 1.5186
hour <= 7.5
samples = 36688
value = -0.0586
hour <= 12.5
samples = 1537
value = -0.6746
dayofweek <= 1.5
samples = 338
value = 1.3674
dayofweek <= 2.5
samples = 74
value = 2.2093
samples = 12818
value = -0.227
samples = 23870
value = 0.0319
samples = 1365
value = -0.534
samples = 172
value = -1.7905
samples = 126
value = 1.8147
samples = 212
value = 1.1016
samples = 34
value = 2.7349
samples = 40
value = 1.7626

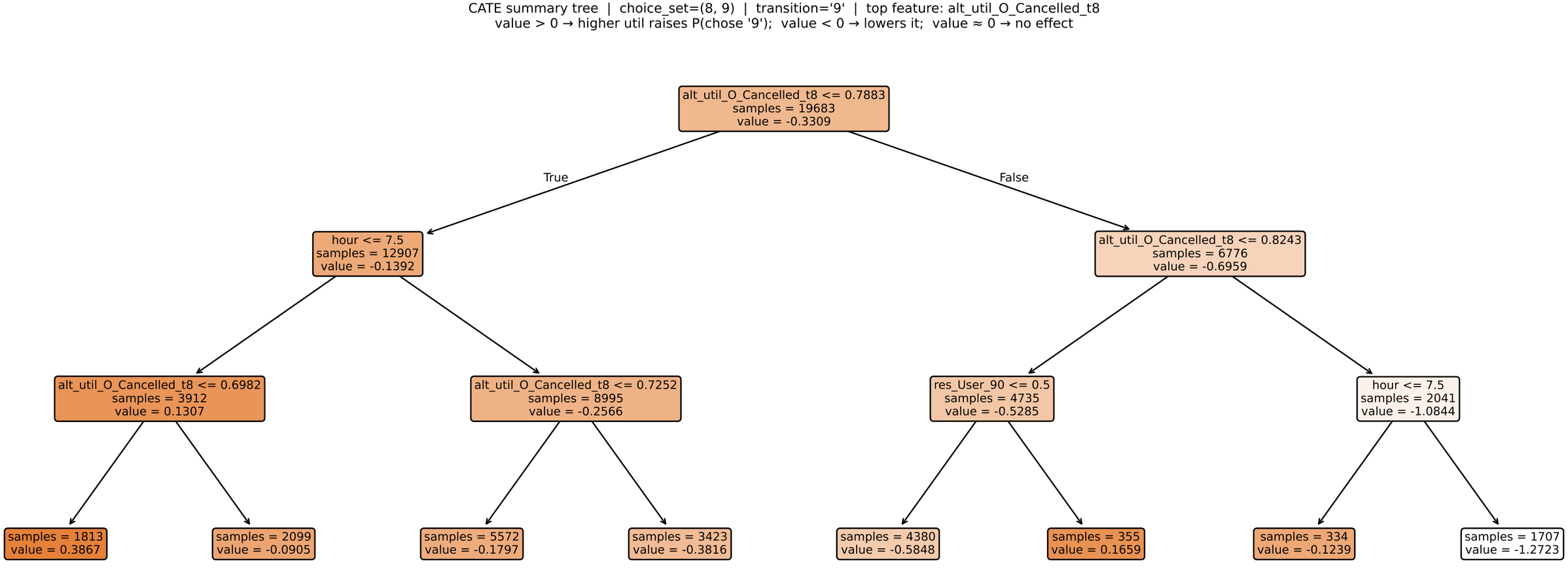

CATE — choice_set=(6, 7) | transition='7' | n=38637 | ATE=-0.2064 std=0.8286
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.2064
Count
CATE θ(X)
Mean CATE by resource
Mean CATE


CATE summary tree | choice_set=(6, 7) | transition='7' | top feature: res_User_107
value > 0 → higher util raises P(chose '7'); value < 0 → lowers it; value ≈ 0 → no effect
res_User_107 <= 0.5
samples = 38637
value = -0.2064
True
False
res_User_2 <= 0.5
samples = 38439
value = -0.1947
alt_util_O_Cancelled_t6 <= 0.6445
samples = 198
value = -2.4852
res_User_1 <= 0.5
samples = 38264
value = -0.1848
alt_util_O_Cancelled_t6 <= 0.7148
samples = 175
value = -2.3643
dayofweek <= 0.5
samples = 43
value = -6.054
alt_util_O_Cancelled_t6 <= 0.7383
samples = 155
value = -1.4951
samples = 28550
value = -0.2456
samples = 9714
value = -0.0061
samples = 29
value = -0.7081
samples = 146
value = -2.6933
samples = 16
value = -3.1276
samples = 27
value = -7.7881
samples = 98
value = -1.0493
samples = 57
value = -2.2617


CATE — choice_set=(8, 9) | transition='8' | n=19683 | ATE=-0.0609 std=0.2901
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.0609
Count
CATE θ(X)
Mean CATE by resource
Mean CATE


CATE summary tree | choice_set=(8, 9) | transition='8' | top feature: alt_util_O_Accepted_t9
value > 0 → higher util raises P(chose '8'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_O_Accepted_t9 <= 0.5625
samples = 19683
value = -0.0609
True
False
alt_util_O_Accepted_t9 <= 0.4375
samples = 6031
value = -0.1663
res_User_128 <= 0.5
samples = 13652
value = -0.0143
hour <= 8.5
samples = 405
value = -0.4362
res_User_100 <= 0.5
samples = 5626
value = -0.1468
hour <= 12.5
samples = 13445
value = -0.0082
hour <= 6.5
samples = 207
value = -0.4118
samples = 151
value = -0.5978
samples = 254
value = -0.3402
samples = 5341
value = -0.1597
samples = 285
value = 0.0948
samples = 9692
value = 0.0232
samples = 3753
value = -0.0895
samples = 182
value = -0.4999
samples = 25
value = 0.2296


CATE — choice_set=(8, 9) | transition='9' | n=19683 | ATE=-0.3309 std=0.6510
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.3309
Count
CATE θ(X)
Mean CATE by resource
Mean CATE


CATE summary tree | choice_set=(8, 9) | transition='9' | top feature: alt_util_O_Cancelled_t8
value > 0 → higher util raises P(chose '9'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_O_Cancelled_t8 <= 0.7883
samples = 19683
value = -0.3309
True
False
hour <= 7.5
samples = 12907
value = -0.1392
alt_util_O_Cancelled_t8 <= 0.8243
samples = 6776
value = -0.6959
alt_util_O_Cancelled_t8 <= 0.6982
samples = 3912
value = 0.1307
alt_util_O_Cancelled_t8 <= 0.7252
samples = 8995
value = -0.2566
res_User_90 <= 0.5
samples = 4735
value = -0.5285
hour <= 7.5
samples = 2041
value = -1.0844
samples = 1813
value = 0.3867
samples = 2099
value = -0.0905
samples = 5572
value = -0.1797
samples = 3423
value = -0.3816
samples = 4380
value = -0.5848
samples = 355
value = 0.1659
samples = 334
value = -0.1239
samples = 1707
value = -1.2723

CATE statistics summary
cate_std = heterogeneity; pct_positive/negative = share of cases where util helps/hurts; top_feature = main moderator

choice_set	transition	n	ate	cate_std	cate_min	cate_p25	cate_p50	cate_p75	cate_max	pct_positive	pct_negative	top_feature
(1, 2, 3)	1	42936	-0.117371	2.524088	-15.530753	-1.459188	-0.033185	1.280956	16.107253	49.3	50.7	alt_util_O_Sent (mail and online)_t3
(1, 2, 3)	2	42936	-0.193329	1.405485	-8.277781	-0.864677	-0.094169	0.601461	8.444389	46.0	54.0	alt_util_O_Sent (mail and online)_t3
(1, 2, 3)	3	42936	-0.065195	2.371381	-12.965909	-1.406123	-0.0585	1.31976	12.75388	48.8	51.2	alt_util_O_Sent (online only)_t1
(4, 5)	4	1908	-0.525971	0.412539	-2.172692	-0.769897	-0.536942	-0.2371	0.405537	10.7	89.3	dayofweek
(4, 5)	5	1908	-0.78197	0.765856	-3.606216	-1.279341	-0.571484	-0.251874	0.988287	10.3	89.7	alt_util_O_Cancelled_t4
(6, 7)	6	38637	-0.066281	0.631645	-4.591133	-0.151499	-6.6e-05	0.077258	4.306787	46.8	53.2	res_User_53
(6, 7)	7	38637	-0.206431	0.828559	-8.177386	-0.339329	-0.005475	0.013476	4.755192	35.4	64.6	res_User_107
(8, 9)	8	19683	-0.06089	0.29007	-0.982644	-0.240051	-0.047232	0.121038	1.357792	43.3	56.7	alt_util_O_Accepted_t9
(8, 9)	9	19683	-0.330853	0.650999	-4.578059	-0.601747	-0.22571	0.055936	1.95187	28.9	71.1	alt_util_O_Cancelled_t8

Key Assumptions & Interpretation Notes

▸ Backdoor criterion (all methods)

X = {hour, dayofweek, n_alternatives, res_*} must block all confounding paths T ← C → Y.

Any variable that affects both util and routing must be in X. Unobserved confounders

cannot be detected and will bias all estimates.

▸ SUTVA (all methods)

One case's routing does not affect another's treatment or outcome.

Violated if resource occupation by one case changes the util seen by a concurrent case.

▸ Positivity (all methods)

Every covariate combination X must have positive probability for all util levels T.

Diagnosed by the covariate balance (SMD) in the backdoor check.

▸ Linearity (Double ML only)

θ is the best-fit linear slope. If the effect varies with X, θ is a variance-weighted

average of CATEs — valid as a summary, but not the simple mean.

Double ML is more powerful than the forest for testing 'is there any effect?'

▸ Honest splitting (Causal Forest only)

Tree structure and leaf estimates use separate subsamples. Yields valid CIs but

halves effective sample size → forest CIs are wider than Double ML's.

A significant Double ML θ with an insignificant forest ATE is not a contradiction.

▸ What the analysis cannot establish

Causal direction (assumes util → choice, not the reverse).

External validity beyond the observed covariate distribution.

Dynamic/lagged effects across cases over time.

Full documentation: ASSUMPTIONS.md